

Qalitrack Manned only. This guide covers Qalitrack Manned. Qalitrack Unmanned has its own installation guide.

WHAT IS IN THE DEPLOYMENT FOLDER

Everything needed to install Qalitrack on a site is contained in the Deployment folder. The folder is handed to the installer as a single package and should not be modified.

install.bat	Double-click to install the backend — the only file you need to run
install.ps1	The PowerShell script called automatically by install.bat
docker-compose.yml	Defines all backend services
.env	Site-specific secrets — provided by the Qalitrack team, never modify
QalitrackHardware.exe	Hardware client installer (weighbridge, camera, RFID, NFC)
QalitrackFrontend.exe	Frontend desktop application installer

PART 1 — BACKEND SERVICES

The backend is a set of services that run inside Docker containers. Docker Desktop and WSL2 are installed automatically by the script if they are not already present on the machine — no manual installation is required. If a restart is needed mid-install, the script will say so clearly; simply restart the PC and double-click **install.bat** again to continue.

Before running the installer, place the **.env** file provided for this site in the same folder as **install.bat**. Then double-click **install.bat** and click **Yes** on the Windows security prompt. The script will clone the source code into WSL2, build all service images, and start every service automatically. The first run takes between five and fifteen minutes depending on the machine's internet speed.

Once the green success banner appears in the terminal window, open a browser and confirm that the following addresses respond: Gateway at **http://localhost:7000**, User Service at **http://localhost:7001**, Master Data at **http://localhost:7002**, Backup at **http://localhost:7003**, and Transactions at **http://localhost:7004**. The backend is configured to restart automatically every time Windows starts, so no further action is needed after a reboot.

To update the backend when a new version is available, re-run **install.bat**. The script pulls the latest code and rebuilds the images while preserving all existing data.

PART 2 — HARDWARE CLIENT

The hardware client collects weight readings from the weighbridge, video from the RTSP camera, and optionally reads from an RFID or NFC device. It runs as a Windows Service and starts automatically on boot.

Right-click **QalitrackHardware.exe** and choose *Run as administrator* — Administrator rights are required to register a Windows Service. The setup wizard will open. Under the Weighbridge section, choose TCP if the weighbridge is connected over the network and enter its IP address and port (the default port is 3002), or choose Serial and select the COM port and baud rate from the weighbridge manual. Under Camera, enter the IP address, RTSP port (typically 554), the stream path, and the login credentials. Tick the RFID and NFC sections only if those readers are physically installed.

Click **Install Service**. The wizard copies the application to `C:\Program Files\Qalitrack`, registers it as a service, and starts it immediately. Verify the installation by opening `http://localhost:5000` in a browser. To update, run the new **QalitrackHardware.exe** as Administrator — it will detect the existing configuration, pre-fill all fields, and the button will read *Update & Reinstall*.

PART 3 — FRONTEND

The frontend is the Qalitrack desktop application. Double-click **QalitrackFrontend.exe**, follow the on-screen steps, and accept the defaults. The installer creates a desktop shortcut when it finishes. To update, run the new **QalitrackFrontend.exe** — it upgrades in place without removing any settings.

TROUBLESHOOTING

Symptom	Fix
install.bat says restart required	Restart the PC and re-run install.bat. Happens once on first install.
Docker not running	Open Docker Desktop, wait for the whale icon, then re-run install.bat.
.env not found	Place .env in the same folder as install.bat and re-run.
WSL not installed	Run <code>wsl --install</code> as Administrator, restart, re-run install.bat.
Services unhealthy	Wait 60 s — databases take time on first boot.
Hardware: Installation Failed	Run QalitrackHardware.exe as Administrator.
Weighbridge — no data (TCP)	Confirm IP and port. Test connectivity: <code>ping <ip></code> .
Weighbridge — no data (Serial)	Confirm COM port in Device Manager and baud rate from the manual.
Camera not connecting	Test in VLC with <code>rtsp://user:pass@ip:554/path</code> . Service retries automatically.